

Biruk Geremew Abza

Software Engineer

✉ birukabza@gmail.com ☎ +251977616186 📍 Addis Ababa, Ethiopia 🌐 birukabza 🌐 linkedin.com/in/birukabza

EDUCATION

BSc in Computer Science

Dec 2022 – Jun 2026 | Addis Ababa, Ethiopia

Addis Ababa University 🌐

- **GPA:** 3.8.
- **Relevant Courses:** Data Structures and Algorithms, Calculus, Discrete Mathematics, Web application Development, Networking, OOP, Advanced Database System, Computer Organizations and Assembly, Intro to Software Engineering.

Advanced Algorithms and Competitive Programming

Apr 2024 – Sep 2024 | Addis Ababa Ethiopia

African To Silicon Valley 🌐

- Solved over **1000** coding challenges across leetcode and codeforces.
- A2SV (Africa to Silicon Valley) is a U.S.-based, **Google-backed** tech academy and incubator that has successfully placed **70+** students in top tech companies.

PROFESSIONAL EXPERIENCE

AI Software Engineer

Jul 2025 – Present | Addis Ababa, Ethiopia

Icog Labs 🌐

- **Architected and developed** emotionally intelligent AI characters for the Hyperon Prototyping project, integrating psychological models with **LLMs** for dynamic, human-like interactions across VR and metaverse platforms.
- **Tripled chat performance** and improved user experience by slashing average response latency from 16s to 5s. This was achieved by implementing **WebSocket-based response streaming**, deferring non-critical animations, and optimizing vector search by **batching FAISS write operations** in a background queue.
- Led a **comprehensive code refactor** of the AI character chat system, replacing multiple fragmented session managers with a **unified, database-driven** session architecture that reduced code complexity and improved maintainability.

Backend Developer Intern

Jan 2025 – Jun 2025 | Addis Ababa, Ethiopia

Icog Labs 🌐

- Contributed to **Rejuve.bio's** 🌐 hypothesis-generation platform — a system that combines cross-species genomic data, LLMs, and symbolic reasoning to accelerate the discovery of anti-aging therapeutics and biomarkers.
- Optimized genomic variant annotation workflows, **cutting runtime by 66%** via concurrent execution and batch processing.
- Implemented real-time task feedback using **WebSocket** between Flask & Prefect services, improving system responsiveness.
- Developed api architecture for multi-provider LLM integration (OpenAI, Anthropic, Gemini), enabling scalable system design for biomedical hypothesis generation.
- Engineered a literature mining engine using PubMed API + Sentence Transformers, boosting LLM accuracy by **40%** for causal gene prediction.

PROJECTS

AgriLo 🌐

Python, Typescript, FastAPI, NextJS

- Led a team of **5 developers** to build AgriLo, an AI-driven platform enabling smallholder farmers to receive real-time, multilingual (6+ languages) guidance via voice/text for crop disease diagnosis, soil analysis, weather-based decisions, and personalized farming plans.
- Secured **1st place** in the **Reboot the Earth National Hackathon 2025** and in the **OpenEPI x UNLEASH Global Hackathon 2025**, delivering impactful, accessible farming solutions for smallholder farmers globally.

MeetMe 🌐

Python, JavaScript, Django, React, Cloundinary

- Architected and built a full-stack social media platform to serve as a portfolio and community hub for artists.
- **Designed a comprehensive RESTful API** with over **25+ endpoints** using Django REST Framework for handling user profiles, artwork uploads, and social interactions.

SKILLS

Programming Languages — Python, JavaScript, C++, SQL, HTML/CSS.

Developer Tools — Git, Docker, Prefect, Firebase, JWT

Frameworks & Libraries — Flask, FastApi, ReactJS, TailwindCSS, Langchain, Autogen, Tensorflow

CERTIFICATES

AWS Certified Solutions Architect– Associate [↗](#) | AWS Certified Cloud Practitioner [↗](#) | Advanced Learning Algorithms [↗](#)